

Appendix A: An Illustrative Example of the Bisection Method

Suppose that $x^2 - 3$ is $f(x)$ and we want to find a solution of $x^2 - 3 = 0$. Let the initial values of x_0 and x_1 be 1 and 2, respectively. Let the predefined tolerable error e be 0.1. In the first loop of the while statement, x_2 is assigned to 1.5 as a result of the division of the sum $x_1 + x_0$ by 2. Since $f(x_0) * f(x_2)$ (that is, $f(1) * f(1.5)$) is positive, x_0 is assigned to 1.5, which implies that the potential root is within the interval $[1.5, 2]$. In the second loop of the while statement, x_2 is calculated again in the same way as before and assigned to 1.75. Since $f(1.5) * f(1.75)$ is negative, x_1 is assigned to 1.75. Thus, the latest interval that contains the potential solution becomes $[1.5, 1.75]$. Execute the while statement again, x_2 is assigned to 1.625 in the third loop. As $f(1.5) * f(1.625)$ is positive, x_0 is assigned to 1.625. This leads to the formation of the latest interval $[1.625, 1.75]$. In the fourth loop, x_2 is assigned to 1.6875. Since $f(1.625) * f(1.6875)$ is positive, x_0 is assigned to 1.6875 and the latest interval $[1.6875, 1.75]$ is formed. Here, $x_1 - x_0 = 0.0625$ is less than e whose value is defined as 0.1. Therefore, the value of 1.6875 is taken as the solution of $f(x) = 0$ in the algorithm. The changes in the values of the related variables in solving equation $x^2 - 3 = 0$ are illustrated in Fig. 1. Note that the tolerable error e is set to 0.1 in this example, but it is possible to make e even smaller to obtain a more accurate solution.

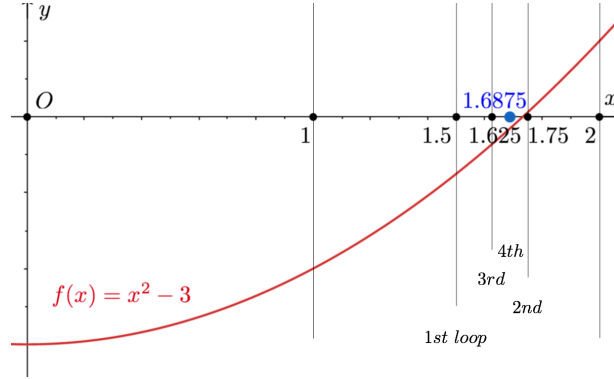


Fig. 1: Changes of variable values in solving the equation $x^2 - 3 = 0$ using the bisection Method

Appendix B: Detailed Results of the Experiment 1

This appendix presents the full details of the comparisons among the testing methods used in Experiment 1. The table includes the following data elements:

- Id: a unique number for each code tested.
- {VM, OVM, FZ, Z3} TDN: the number of test data generated using the V-Method, the original V-Method, fuzzing, or the SMT solver Z3.
- {VM, OVM, FZ, Z3} BC: the branch coverage achieved using the V-Method, the original V-Method, fuzzing, or the SMT solver Z3.
- {VM, OVM, FZ, Z3} IC: the instruction coverage achieved using the V-Method, the original V-Method, fuzzing, or the SMT solver Z3.
- LOC: the number of lines in the program tested.

TABLE I: Detailed comparison of V-Method (VM), original V-Method (OVM), Z3, and fuzzing (FZ) on all 152 programs.

ID	TDN	VM BC	VM IC	OVM BC	OVM IC	Z3 BC	Z3 IC	FZ BC	FZ IC	LOC
1	8	100.0%	98.7%	85.7%	98.7%	100.0%	98.7%	64.3%	92.5%	61
2	5	100.0%	97.9%	75.0%	97.9%	100.0%	97.9%	87.5%	97.9%	58
3	4	100.0%	94.5%	83.3%	94.5%	83.3%	94.5%	66.7%	65.1%	78
4	3	100.0%	91.7%	100.0%	91.7%	100.0%	91.7%	75.0%	91.7%	37
5	3	100.0%	95.2%	100.0%	95.2%	100.0%	95.2%	75.0%	90.5%	61
6	3	100.0%	95.2%	83.3%	95.2%	83.3%	95.2%	75.0%	95.2%	105
7	5	100.0%	97.4%	100.0%	97.4%	75.0%	97.4%	87.5%	97.4%	124
8	8	100.0%	98.9%	75.0%	81.5%	58.3%	60.7%	50.0%	68.1%	120
9	5	100.0%	98.0%	100.0%	98.0%	81.2%	95.4%	81.2%	95.4%	80
10	4	100.0%	98.8%	90.0%	98.0%	80.0%	98.0%	90.0%	95.5%	93
11	5	100.0%	98.7%	90.0%	94.8%	80.0%	90.0%	80.0%	96.1%	148
12	8	100.0%	98.8%	80.0%	74.1%	50.0%	57.9%	80.0%	74.1%	123
13	100	90.0%	99.0%	80.0%	99.0%	90.0%	99.0%	70.0%	73.1%	503
14	3	100.0%	99.1%	100.0%	99.1%	100.0%	99.1%	100.0%	99.1%	191

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ID	TDN	VM BC	VM IC	OVM BC	OVM IC	Z3 BC	Z3 IC	FZ BC	FZ IC	LOC
15	10	100.0%	99.0%	77.8%	85.5%	55.6%	76.6%	27.8%	40.9%	139
16	8	100.0%	95.7%	87.5%	95.7%	37.5%	57.5%	62.5%	65.3%	114
17	100	66.7%	94.5%	66.7%	94.6%	50.0%	89.2%	50.0%	91.0%	469
18	4	100.0%	98.4%	92.9%	97.3%	35.7%	67.2%	35.7%	67.2%	102
19	4	100.0%	95.3%	100.0%	95.3%	100.0%	95.3%	75.0%	64.1%	48
20	5	100.0%	98.5%	100.0%	98.5%	85.7%	96.0%	92.9%	97.5%	107
21	4	100.0%	98.6%	77.8%	89.6%	55.6%	70.6%	83.3%	82.4%	117
22	5	100.0%	98.6%	100.0%	98.6%	100.0%	98.6%	75.0%	80.6%	84
23	4	100.0%	98.5%	100.0%	98.5%	100.0%	98.5%	33.3%	48.3%	80
24	4	100.0%	98.4%	100.0%	98.4%	100.0%	98.4%	66.7%	89.6%	94
25	100	87.5%	99.4%	62.5%	95.5%	62.5%	95.5%	75.0%	99.4%	455
26	100	87.5%	99.4%	62.5%	95.5%	62.5%	95.5%	62.5%	95.5%	453
27	100	66.7%	94.4%	66.7%	94.4%	66.7%	94.4%	83.3%	99.2%	454
28	5	100.0%	97.2%	83.3%	94.5%	83.3%	94.5%	83.3%	97.2%	74
29	5	100.0%	97.2%	83.3%	94.5%	83.3%	94.5%	66.7%	93.6%	74
30	100	83.3%	98.5%	83.3%	98.5%	83.3%	98.5%	83.2%	98.2%	453
31	6	100.0%	97.3%	50.0%	77.3%	83.3%	94.6%	83.3%	97.3%	80
32	100	83.3%	99.2%	50.0%	94.4%	50.0%	94.4%	66.7%	98.2%	454
33	3	100.0%	97.1%	16.7%	69.3%	16.7%	69.3%	83.3%	97.1%	67
34	5	100.0%	97.2%	50.0%	76.6%	33.3%	72.9%	83.3%	97.2%	74
35	6	100.0%	97.9%	33.3%	54.6%	16.7%	52.5%	66.7%	68.5%	89
36	5	100.0%	98.0%	50.0%	82.8%	33.3%	82.8%	50.0%	67.3%	87
37	5	100.0%	97.2%	50.0%	76.6%	33.3%	72.9%	50.0%	89.0%	76
38	5	100.0%	97.2%	50.0%	75.7%	50.0%	76.6%	83.3%	97.2%	74
39	100	75.0%	96.6%	75.0%	96.6%	75.0%	96.6%	50.0%	96.6%	467
40	8	100.0%	98.6%	100.0%	98.6%	50.0%	85.6%	100.0%	74.9%	104
41	5	100.0%	97.2%	66.7%	79.4%	50.0%	76.6%	66.7%	93.6%	74
42	5	100.0%	97.7%	80.0%	84.5%	60.0%	79.6%	80.0%	95.8%	86
43	100	87.5%	97.7%	75.0%	97.3%	75.0%	97.3%	87.5%	87.1%	469
44	100	80.0%	95.5%	80.0%	95.5%	80.0%	95.5%	60.0%	82.9%	475
45	6	100.0%	99.0%	66.7%	89.7%	33.3%	67.3%	66.7%	72.4%	96
46	5	100.0%	97.7%	75.0%	83.1%	62.5%	80.0%	87.5%	95.5%	89
47	5	100.0%	97.7%	100.0%	97.7%	100.0%	97.7%	87.5%	95.5%	89
48	5	100.0%	97.8%	100.0%	97.8%	100.0%	97.8%	83.3%	94.2%	91
49	5	100.0%	98.0%	100.0%	98.0%	100.0%	98.0%	75.0%	90.1%	93
50	5	100.0%	97.9%	100.0%	97.9%	100.0%	97.9%	70.0%	89.7%	91
51	6	100.0%	98.4%	100.0%	98.4%	100.0%	98.4%	71.4%	92.2%	103
52	5	100.0%	97.8%	100.0%	97.8%	100.0%	97.8%	80.0%	95.7%	89
53	5	100.0%	97.9%	100.0%	97.9%	100.0%	97.9%	80.0%	89.7%	90
54	5	100.0%	97.7%	100.0%	97.7%	100.0%	97.7%	87.5%	95.5%	89
55	5	100.0%	97.7%	100.0%	97.7%	100.0%	97.7%	75.0%	91.7%	87
56	4	100.0%	97.8%	100.0%	97.8%	100.0%	97.8%	80.0%	92.6%	85
57	100	75.0%	95.7%	75.0%	95.7%	75.0%	95.7%	100.0%	91.9%	466
58	6	100.0%	98.7%	100.0%	98.7%	100.0%	98.7%	75.0%	64.0%	90
59	5	100.0%	98.9%	100.0%	98.9%	100.0%	98.9%	66.7%	60.5%	88
60	6	100.0%	98.8%	100.0%	98.8%	100.0%	98.8%	100.0%	98.8%	90
61	4	100.0%	97.2%	100.0%	97.2%	100.0%	97.2%	75.0%	87.9%	73
62	5	100.0%	96.6%	100.0%	96.6%	100.0%	96.6%	50.0%	78.0%	73
63	4	100.0%	96.6%	100.0%	96.6%	100.0%	96.6%	83.3%	93.3%	65
64	4	100.0%	96.6%	100.0%	96.6%	100.0%	96.6%	33.3%	83.1%	65
65	5	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	83.3%	96.7%	69
66	4	100.0%	96.9%	100.0%	96.9%	100.0%	96.9%	75.0%	90.8%	61
67	4	100.0%	96.6%	100.0%	96.6%	100.0%	96.6%	83.3%	96.6%	65
68	4	100.0%	96.6%	100.0%	96.6%	100.0%	96.6%	33.3%	83.1%	65
69	3	100.0%	95.9%	100.0%	95.9%	100.0%	95.9%	50.0%	86.5%	51
70	5	100.0%	93.8%	100.0%	93.8%	100.0%	93.8%	66.7%	86.6%	68
71	6	100.0%	95.5%	100.0%	95.5%	100.0%	95.5%	50.0%	71.6%	78
72	3	100.0%	95.0%	100.0%	95.0%	100.0%	95.0%	100.0%	95.0%	67
73	6	100.0%	96.1%	100.0%	96.1%	100.0%	96.1%	80.0%	94.1%	88
74	8	100.0%	96.4%	100.0%	96.4%	100.0%	96.4%	40.0%	62.4%	103
75	5	100.0%	93.2%	100.0%	93.2%	100.0%	93.2%	75.0%	87.9%	81
76	6	100.0%	95.7%	100.0%	95.7%	100.0%	95.7%	66.7%	86.4%	80
77	3	100.0%	94.5%	100.0%	94.5%	100.0%	94.5%	75.0%	89.1%	71
78	3	100.0%	94.6%	100.0%	94.6%	100.0%	94.6%	75.0%	91.9%	68
79	3	100.0%	94.8%	100.0%	94.8%	100.0%	94.8%	25.0%	71.6%	69
80	3	100.0%	97.6%	100.0%	97.6%	100.0%	97.6%	83.3%	90.3%	76
81	3	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	87.5%	96.7%	66
82	3	100.0%	94.5%	100.0%	94.5%	100.0%	94.5%	80.0%	89.0%	74
83	3	100.0%	95.8%	100.0%	95.8%	100.0%	95.8%	75.0%	91.5%	54
84	3	100.0%	93.9%	100.0%	93.9%	100.0%	93.9%	87.5%	90.9%	70
85	3	100.0%	95.8%	100.0%	95.8%	100.0%	95.8%	75.0%	91.5%	55
86	100	75.0%	98.5%	75.0%	98.5%	75.0%	98.5%	75.0%	98.5%	458
87	7	100.0%	93.6%	100.0%	93.6%	100.0%	93.6%	75.0%	75.0%	90
88	3	100.0%	92.9%	100.0%	92.9%	100.0%	92.9%	100.0%	92.9%	66
89	5	100.0%	95.7%	100.0%	95.7%	100.0%	95.7%	50.0%	78.7%	77
90	3	100.0%	94.7%	100.0%	94.7%	100.0%	94.7%	75.0%	88.6%	74

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ID	TDN	VM BC	VM IC	OVM BC	OVM IC	Z3 BC	Z3 IC	FZ BC	FZ IC	LOC
91	3	100.0%	92.0%	100.0%	92.0%	100.0%	92.0%	75.0%	83.9%	80
92	3	100.0%	92.0%	100.0%	92.0%	100.0%	92.0%	75.0%	83.9%	80
93	3	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	87.5%	96.7%	67
94	3	100.0%	94.9%	100.0%	94.9%	100.0%	94.9%	75.0%	82.2%	73
95	3	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	87.5%	96.7%	65
96	3	100.0%	94.6%	100.0%	94.6%	100.0%	94.6%	50.0%	87.5%	51
97	3	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	67
98	3	100.0%	96.8%	100.0%	96.8%	100.0%	96.8%	50.0%	78.9%	65
99	3	100.0%	96.6%	100.0%	96.6%	100.0%	96.6%	50.0%	84.3%	64
100	3	100.0%	94.6%	100.0%	94.6%	100.0%	94.6%	25.0%	72.3%	71
101	3	100.0%	94.7%	100.0%	94.7%	100.0%	94.7%	50.0%	89.5%	50
102	3	100.0%	94.5%	100.0%	94.5%	100.0%	94.5%	50.0%	81.8%	71
103	3	100.0%	94.7%	100.0%	94.7%	100.0%	94.7%	100.0%	94.7%	50
104	3	100.0%	94.7%	100.0%	94.7%	100.0%	94.7%	50.0%	87.7%	50
105	3	100.0%	94.6%	100.0%	94.6%	100.0%	94.6%	100.0%	94.6%	51
106	3	100.0%	94.7%	100.0%	94.7%	100.0%	94.7%	50.0%	89.5%	50
107	3	100.0%	97.2%	100.0%	97.2%	100.0%	97.2%	25.0%	74.3%	65
108	3	100.0%	92.1%	100.0%	92.1%	100.0%	92.1%	100.0%	92.1%	65
109	3	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	37.5%	82.6%	65
110	3	100.0%	96.7%	100.0%	96.7%	100.0%	96.7%	62.5%	85.9%	67
111	3	100.0%	94.5%	100.0%	94.5%	100.0%	94.5%	75.0%	87.2%	71
112	3	100.0%	97.6%	100.0%	97.6%	100.0%	97.6%	83.3%	90.3%	74
113	3	100.0%	96.9%	100.0%	96.9%	100.0%	96.9%	37.5%	79.2%	67
114	4	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	66.7%	75.0%	1018
115	100	66.7%	92.3%	66.7%	92.3%	66.7%	92.3%	33.3%	46.2%	938
116	100	92.9%	98.8%	92.9%	98.8%	92.9%	98.8%	92.9%	98.8%	1025
117	100	90.0%	97.6%	90.0%	96.4%	90.0%	96.4%	70.0%	89.2%	947
118	100	82.1%	92.3%	82.1%	92.3%	82.1%	92.3%	78.6%	92.3%	842
119	100	83.3%	96.5%	83.3%	96.5%	83.3%	96.5%	83.3%	96.5%	736
120	100	70.0%	83.4%	70.0%	83.4%	57.5%	52.5%	52.5%	52.5%	1147
121	100	65.0%	91.8%	45.0%	59.6%	62.5%	90.7%	40.0%	55.5%	1013
122	100	64.1%	89.3%	59.8%	80.9%	59.8%	80.9%	63.0%	81.8%	1078
123	100	62.5%	85.3%	62.5%	85.3%	37.5%	59.6%	37.5%	59.6%	977
124	100	62.5%	89.0%	62.5%	89.0%	37.5%	59.1%	37.5%	59.1%	741
125	100	85.7%	99.4%	85.7%	99.4%	50.0%	82.2%	50.0%	91.9%	1012
126	100	70.0%	79.4%	70.0%	79.4%	66.7%	78.0%	70.0%	79.4%	893
127	4	100.0%	93.5%	100.0%	93.5%	100.0%	93.5%	75.0%	91.3%	1008
128	5	100.0%	90.3%	100.0%	90.3%	100.0%	90.3%	100.0%	90.3%	951
129	4	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	75.0%	88.8%	979
130	4	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	50.0%	85.9%	945
131	4	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	50.0%	90.4%	1003
132	3	100.0%	100.0%	75.0%	93.5%	50.0%	82.6%	50.0%	84.8%	907
133	20	90.6%	97.4%	90.6%	97.4%	62.5%	73.8%	50.0%	60.6%	1920
134	20	94.4%	74.8%	94.4%	73.6%	72.2%	61.1%	61.1%	60.0%	1166
135	20	97.2%	98.5%	97.2%	98.1%	91.7%	97.3%	30.6%	62.8%	719
136	20	95.0%	94.2%	95.0%	94.2%	75.0%	92.6%	15.0%	20.5%	622
137	11	100.0%	97.2%	100.0%	97.2%	81.2%	93.9%	50.0%	68.2%	513
138	20	25.0%	20.7%	20.7%	12.3%	25.0%	19.2%	17.6%	14.0%	1056
139	20	83.3%	93.3%	58.3%	93.3%	81.8%	92.4%	66.7%	83.3%	1002
140	20	100.0%	88.2%	100.0%	87.0%	100.0%	88.2%	50.0%	76.4%	680
141	20	100.0%	90.2%	100.0%	90.2%	87.5%	86.0%	87.5%	86.0%	745
142	20	75.0%	81.9%	75.0%	81.9%	66.7%	81.9%	66.7%	81.9%	731
143	11	100.0%	95.8%	100.0%	95.8%	70.0%	81.7%	60.0%	74.2%	454
144	16	100.0%	97.3%	100.0%	97.3%	71.4%	85.7%	71.4%	83.0%	581
145	17	100.0%	97.0%	75.0%	97.0%	50.0%	81.8%	50.0%	81.2%	583
146	20	28.3%	30.3%	24.0%	30.3%	28.3%	30.3%	13.3%	12.8%	1074
147	20	100.0%	92.7%	100.0%	92.7%	100.0%	89.9%	78.6%	81.0%	939
148	20	62.5%	90.2%	62.5%	90.2%	60.4%	89.6%	58.3%	89.1%	1601
149	10	100.0%	100.0%	100.0%	99.6%	100.0%	100.0%	75.0%	82.4%	520
150	20	90.0%	93.8%	80.0%	93.8%	72.0%	83.2%	22.0%	19.2%	1012
151	20	93.8%	94.8%	93.8%	94.8%	81.2%	92.2%	62.5%	73.3%	1236
152	8	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	91.7%	97.2%	503

To help readers better understand the detailed performance of the four testing methods used in our experiment, we also employ line charts to visualize method comparisons across different contexts. Fig. 2 and Fig. 3 illustrate the comparisons of branch coverage and instruction coverage between the V-Method and other baselines on all 152 programs, respectively.

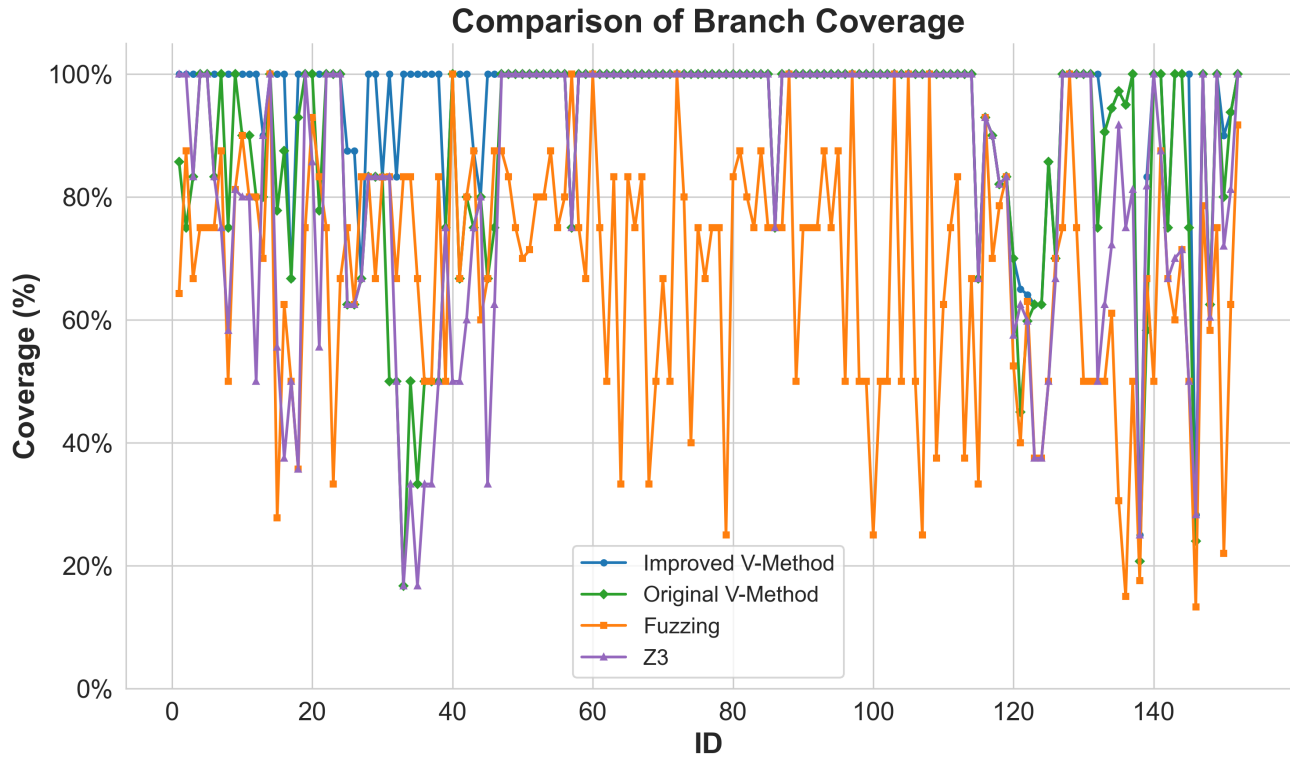


Fig. 2: Comparison of branch coverage for V-Method, original V-Method, fuzzing, and Z3 on all 152 programs.

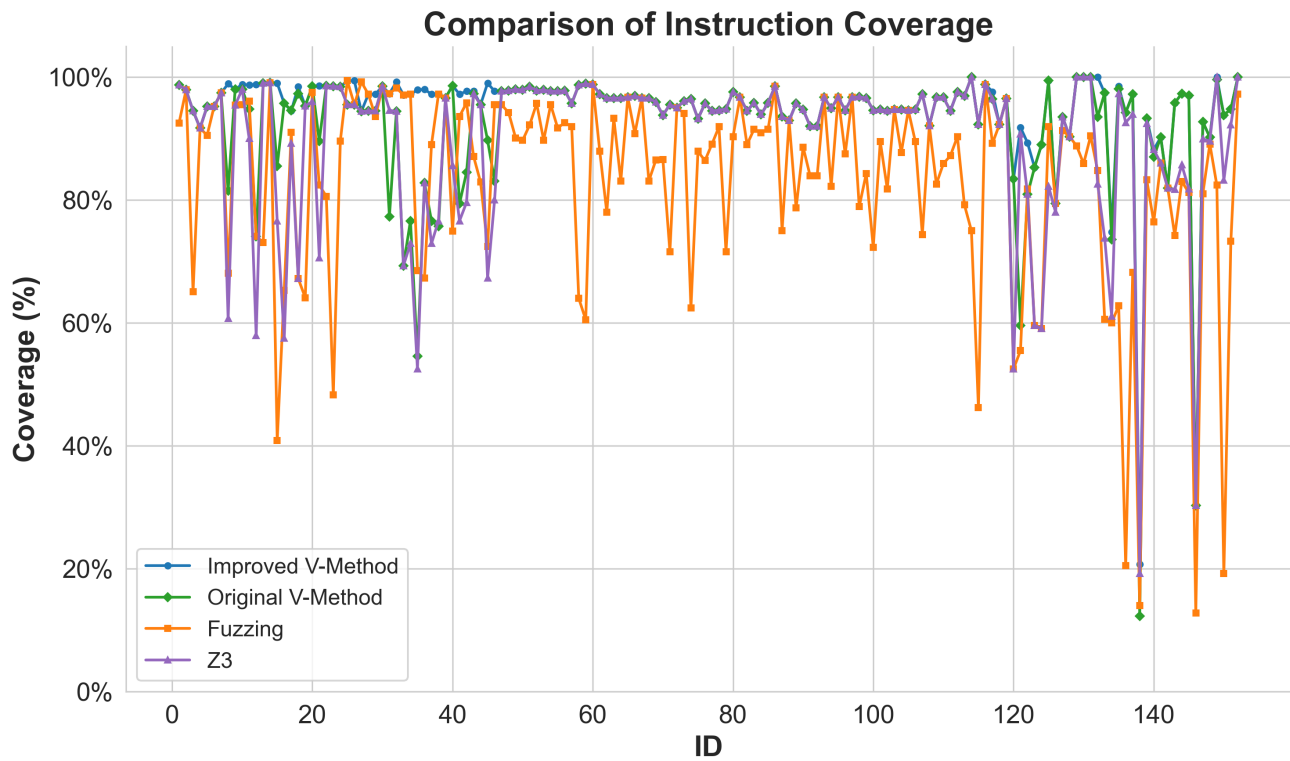


Fig. 3: Comparison of instruction coverage for V-Method, original V-Method, fuzzing, and Z3 on all 152 programs.

Appendix C: Statistical comparison of V-Method (VM) and the baseline methods original V-Method (OVM), Z3, and fuzzing (FZ) across application domains in Experiment 1

TABLE II: Statistical comparison of VM and the baseline methods by application domain in Experiment 1

Application domain	Method	N	Mean TDN	Branch coverage		Instruction coverage	
				Mean (SD)	$p; \delta$	Mean (SD)	$p; \delta$
Financial and Banking Systems	VM	8	4.75	100.00% (0.00%)	—; —	96.50% (1.79%)	—; —
	OVM	8	4.75	92.29% (10.88%)	$1.09 \times 10^{-1}; 0.375$	93.41% (7.95%)	$6.55 \times 10^{-1}; 0.219$
	Z3	8	4.75	89.88% (17.58%)	$1.09 \times 10^{-1}; 0.375$	91.07% (13.46%)	$2.85 \times 10^{-1}; 0.297$
	FZ	8	4.75	72.14% (15.83%)	$7.81 \times 10^{-3}; 1.000$	84.13% (12.28%)	$2.77 \times 10^{-2}; 0.625$
Health and Fitness Monitoring	VM	14	3.93	100.00% (0.00%)	—; —	95.59% (1.85%)	—; —
	OVM	14	3.93	99.29% (2.67%)	$3.17 \times 10^{-1}; 0.071$	95.31% (1.62%)	$3.17 \times 10^{-1}; 0.082$
	Z3	14	3.93	98.57% (5.35%)	$3.17 \times 10^{-1}; 0.071$	94.97% (2.16%)	$3.17 \times 10^{-1}; 0.133$
	FZ	14	3.93	70.36% (19.73%)	$1.21 \times 10^{-3}; 0.929$	86.02% (10.37%)	$2.21 \times 10^{-3}; 0.719$
Home and Building Automation	VM	31	16.48	96.73% (9.56%)	—; —	96.37% (2.40%)	—; —
	OVM	31	16.48	90.41% (17.81%)	$2.77 \times 10^{-2}; 0.169$	93.18% (8.93%)	$2.80 \times 10^{-2}; 0.221$
	Z3	31	16.48	87.32% (24.03%)	$1.76 \times 10^{-2}; 0.180$	92.00% (10.64%)	$1.80 \times 10^{-2}; 0.239$
	FZ	31	16.48	70.33% (17.88%)	$3.40 \times 10^{-6}; 0.822$	87.43% (8.86%)	$1.82 \times 10^{-5}; 0.739$
Industrial and Energy Optimization	VM	27	32.48	94.46% (9.75%)	—; —	96.34% (3.29%)	—; —
	OVM	27	32.48	82.61% (22.14%)	$1.12 \times 10^{-2}; 0.305$	92.29% (7.99%)	$7.63 \times 10^{-3}; 0.339$
	Z3	27	32.48	82.02% (23.22%)	$7.58 \times 10^{-3}; 0.303$	91.64% (10.75%)	$5.01 \times 10^{-3}; 0.326$
	FZ	27	32.48	72.28% (19.06%)	$3.70 \times 10^{-5}; 0.743$	88.78% (13.57%)	$9.99 \times 10^{-4}; 0.439$
Online Management and Education	VM	11	22.91	95.00% (11.18%)	—; —	95.67% (5.90%)	—; —
	OVM	11	22.91	88.12% (10.98%)	$4.31 \times 10^{-2}; 0.372$	94.37% (6.46%)	$1.09 \times 10^{-1}; 0.231$
	Z3	11	22.91	81.65% (21.41%)	$4.31 \times 10^{-2}; 0.372$	89.96% (13.57%)	$6.79 \times 10^{-2}; 0.306$
	FZ	11	22.91	69.52% (18.03%)	$1.13 \times 10^{-2}; 0.802$	79.80% (19.15%)	$1.80 \times 10^{-2}; 0.645$
Transportation and Logistics Management	VM	22	21.82	95.11% (11.64%)	—; —	95.98% (3.33%)	—; —
	OVM	22	21.82	87.15% (16.34%)	$1.78 \times 10^{-2}; 0.279$	92.67% (6.39%)	$1.80 \times 10^{-2}; 0.364$
	Z3	22	21.82	77.00% (26.59%)	$5.01 \times 10^{-3}; 0.378$	86.05% (14.38%)	$5.06 \times 10^{-3}; 0.500$
	FZ	22	21.82	61.90% (18.94%)	$8.28 \times 10^{-5}; 0.849$	83.33% (10.94%)	$5.95 \times 10^{-5}; 0.845$
Warehouse and Inventory Management	VM	19	34.63	92.93% (12.00%)	—; —	96.40% (2.65%)	—; —
	OVM	19	34.63	89.60% (17.96%)	$1.09 \times 10^{-1}; 0.036$	94.45% (8.77%)	$1.80 \times 10^{-1}; 0.089$
	Z3	19	34.63	85.22% (20.02%)	$4.31 \times 10^{-2}; 0.172$	94.50% (4.53%)	$7.96 \times 10^{-2}; 0.269$
	FZ	19	34.63	72.04% (23.05%)	$2.25 \times 10^{-3}; 0.565$	85.10% (14.38%)	$5.30 \times 10^{-4}; 0.745$
Open-source programs	VM	20	17.65	86.75% (22.78%)	—; —	86.42% (21.76%)	—; —
	OVM	20	17.65	83.32% (24.59%)	$4.22 \times 10^{-2}; 0.090$	85.84% (23.14%)	$2.69 \times 10^{-2}; 0.018$
	Z3	20	17.65	73.85% (21.56%)	$9.82 \times 10^{-4}; 0.435$	81.04% (21.33%)	$6.55 \times 10^{-4}; 0.370$
	FZ	20	17.65	53.90% (23.54%)	$1.91 \times 10^{-6}; 0.765$	65.36% (26.72%)	$1.32 \times 10^{-4}; 0.700$
All programs	VM	152	21.21	94.65% (12.56%)	—; —	94.89% (8.87%)	—; —
	OVM	152	21.21	88.27% (18.32%)	$7.35 \times 10^{-8}; 0.196$	92.44% (11.07%)	$1.85 \times 10^{-7}; 0.199$
	Z3	152	21.21	83.61% (22.47%)	$1.59 \times 10^{-10}; 0.262$	90.02% (13.07%)	$1.30 \times 10^{-10}; 0.293$
	FZ	152	21.21	67.55% (20.38%)	$1.83 \times 10^{-23}; 0.779$	83.02% (16.51%)	$1.98 \times 10^{-21}; 0.652$

Notes: BC denotes branch coverage; IC denotes instruction coverage; TDN denotes the number of test data generated. The reported p -values are obtained from paired Wilcoxon signed-rank tests comparing VM with each baseline method. δ denotes Cliff's delta.

Appendix D: Detailed Results of Experiment 2 on the Industrial Railway Interlocking System

This appendix presents the per-program mutation-testing results of Experiment 2 for the 50 programs from the industrial railway interlocking system, comparing the V-Method (VM), original V-Method (OVM), Z3, and fuzzing (FZ).

TABLE III: Per-program mutation-testing results across 50 Java programs.

ID	Program	LOC	Tests	Mut.	VM kill	VM MS	OVM kill	OVM MS	Z3 kill	Z3 MS	FZ kill	FZ MS
i1	i1_PointPositionAndBlockCheck	147	30	50	36	72.00%	33	66.00%	32	64.00%	26	52.00%
i2	i2_PointOperation	534	30	131	104	79.39%	92	70.23%	84	64.12%	65	49.62%
i3	i3_NormalTrainPassDetection	412	30	102	78	76.47%	71	69.61%	63	61.76%	50	49.02%
i4	i4_NormalTrainPassUnlock	487	30	168	148	88.10%	130	77.38%	121	72.02%	89	52.98%
i5	i5_ManualDelayedRouteUnlock	682	30	301	226	75.08%	172	57.14%	153	50.83%	104	34.55%
i6	i6_CancelUnlock	388	30	112	90	80.36%	80	71.43%	75	66.96%	60	53.57%
i7	i7_SectionFaultUnlock	237	30	79	55	69.62%	49	62.03%	44	55.70%	32	40.51%
i8	i8_TrainRouteSet	111	30	15	8	53.33%	8	53.33%	8	53.33%	5	33.33%
i9	i9_TrainRouteLock	571	30	145	122	84.14%	107	73.79%	104	71.72%	83	57.24%
i10	i10_SignalChangeAndTrainRouteUnlock	90	30	17	8	47.06%	5	29.41%	4	23.53%	2	11.76%
i11	i11_SplitPointRouteLockConditionCheck	304	30	60	51	85.00%	46	76.67%	42	70.00%	36	60.00%
i12	i12_RouteSettingPointHandling	110	30	18	11	61.11%	8	44.44%	7	38.89%	5	27.78%
i13	i13_TrainRouteClaim	461	30	149	121	81.21%	106	71.14%	91	61.07%	63	42.28%
i14	i14_SplitPointRouteSetConditionCheck	343	30	98	72	73.47%	62	63.27%	55	56.12%	48	48.98%
i15	i15_TrainSignalChange	835	30	169	150	88.76%	124	73.37%	100	59.17%	75	44.38%
i16	i16_TrainApproachLockCheck	268	30	59	45	76.27%	39	66.10%	36	61.02%	28	47.46%
i17	i17_TrainRouteUnlock	96	30	14	8	57.14%	5	35.71%	4	28.57%	2	14.29%
i18	i18_SplitPointSignalOpenCheck	239	30	59	40	67.80%	35	59.32%	32	54.24%	27	45.76%
i19	i19_SplitPointUnlock	86	30	13	9	69.23%	5	38.46%	5	38.46%	3	23.08%
i20	i20_PointControlDecom	506	30	80	70	87.50%	62	77.50%	59	73.75%	45	56.25%
i21	i21_PointSingleOperation	495	30	66	54	81.82%	46	69.70%	42	63.64%	29	43.94%
i22	i22_RouteSettingOperation	606	30	69	55	79.71%	48	69.57%	45	65.22%	34	49.28%
i23	i23_SwitchNormalOperation	306	30	57	39	68.42%	34	59.65%	30	52.63%	26	45.61%
i24	i24_SwitchPositionCheck	236	30	53	41	77.36%	36	67.92%	36	67.92%	27	50.94%
i25	i25_SwitchReverseOperation	306	30	51	38	74.51%	34	66.67%	30	58.82%	23	45.10%
i26	i26_SwitchOpResult	154	30	37	24	64.86%	21	56.76%	19	51.35%	13	35.14%
i27	i27_RouteLockingCheck	268	30	48	30	62.50%	29	60.42%	26	54.17%	18	37.50%
i28	i28_SignalOpeningPermissionCheck	222	30	52	34	65.38%	29	55.77%	27	51.92%	20	38.46%
i29	i29_SignalKeepingCheck	227	30	59	39	66.10%	33	55.93%	30	50.85%	23	38.98%
i30	i30_RouteReleaseCheck	222	30	51	37	72.55%	32	62.75%	30	58.82%	22	43.14%
i31	i31_AbnormalRouteReleaseCheck	228	30	53	38	71.70%	33	62.26%	29	54.72%	26	49.06%
i32	i32_ManualSwitchOperationCheck	225	30	58	42	72.41%	37	63.79%	33	56.90%	25	43.10%
i33	i33_CallOnRouteLockingCheck	243	30	61	51	83.61%	46	75.41%	42	68.85%	35	57.38%
i34	i34_SignalClosureCheck	205	30	47	38	80.85%	27	57.45%	23	48.94%	17	36.17%
i35	i35_SpecialShuntingRouteCheck	249	30	61	50	81.97%	45	73.77%	41	67.21%	34	55.74%
i36	i36_TrackOccupancyProtectionCheck	226	30	56	36	64.29%	32	57.14%	30	53.57%	21	37.50%
i37	i37_ConflictingRoutePreventionCheck	240	30	69	43	62.32%	41	59.42%	39	56.52%	30	43.48%
i38	i38_PointPositionConsistencyCheck	253	30	75	46	61.33%	43	57.33%	40	53.33%	31	41.33%
i39	i39_RouteIdleStateCheck	242	30	73	50	68.49%	45	61.64%	41	56.16%	33	45.21%
i40	i40_RouteSettingReadinessCheck	248	30	65	47	72.31%	41	63.08%	38	58.46%	25	38.46%
i41	i41_SignalIndicationConsistencyCheck	253	30	75	51	68.00%	43	57.33%	42	56.00%	31	41.33%
i42	i42_RouteCompletionStateCheck	247	30	71	52	73.24%	45	63.38%	40	56.34%	30	42.25%
i43	i43_HazardSignalPermissivePreventionCheck	310	30	80	61	76.25%	55	68.75%	53	66.25%	44	55.00%
i44	i44_ConflictingRouteHazardDetectionCheck	321	30	61	49	80.33%	42	68.85%	38	62.30%	27	44.26%
i45	i45_PointNoIndicationHazardDetectionCheck	321	30	62	50	80.65%	45	72.58%	41	66.13%	32	51.61%
i46	i46_ObstacleOnRouteHazardDetectionCheck	323	30	71	55	77.46%	48	67.61%	42	59.15%	37	52.11%
i47	i47_TrackOccupiedHazardDetectionCheck	323	30	69	54	78.26%	48	69.57%	43	62.32%	34	49.28%
i48	i48_UnsafeRouteReleaseHazardDetectionCheck	298	30	60	48	80.00%	42	70.00%	39	65.00%	32	53.33%
i49	i49_UnsafePointMovementHazardDetectionCheck	287	30	55	47	85.45%	42	76.36%	38	69.09%	28	50.91%
i50	i50_UnsafeSignalOpeningCommandHazardDetection	287	30	58	44	75.86%	42	72.41%	39	67.24%	29	50.00%